

Toponym Recognition Using BERT Models: A Comparative Study and Interactive System

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Abstract

Named Entity Recognition (NER) is a cornerstone of natural language processing (NLP), enabling systems to identify structured information from unstructured text. Among its sub-tasks, toponym recognition — the detection of geographic place names — is critical for applications spanning geospatial intelligence, crisis mapping, historical document analysis, and news geo-tagging. This paper presents a comprehensive comparative study of three state-of-the-art transformer architectures — **BERT**, **RoBERTa**, and **DistilBERT** — each independently fine-tuned on a multi-domain, 187,245-token corpus for toponym-focused NER.

We introduce a full-stack deployment platform comprising a Flask REST API backend and a React-based frontend that enables real-time, side-by-side model comparison on arbitrary input text. The system supports multi-modal input — direct text entry, URL extraction, and document upload (PDF, DOCX, TXT) — and visualises entity spans with colour-coded labels and confidence scores.

Experimental evaluation on a held-out test set across four distinct text domains shows RoBERTa achieving the highest average F1-score of **88.1 ± 0.3** , BERT reaching **85.7 ± 0.4** , and DistilBERT attaining **83.8 ± 0.5** while running **59% faster** on CPU (198 ± 15 ms vs. 487 ± 23 ms for BERT). Our models and interactive system are released as open-source.

Keywords: *Named Entity Recognition, Toponym Recognition, BERT, RoBERTa, DistilBERT, Transfer Learning, Geospatial NLP, Transformer Models, Information Extraction*

1. Introduction

1.1 Background and Motivation

The exponential growth of digitized text — spanning news archives, social media, scientific publications, and historical records — has created an urgent need for automated systems capable of extracting and structuring geographic information. Toponyms, or geographic place names (e.g., cities, countries, rivers, mountains), appear pervasively across all text domains but present particular challenges for automated extraction: they are phonologically and orthographically identical to personal names ("Jordan", "Virginia"), organization names ("Amazon"), and common nouns, making context-sensitive disambiguation essential.

Traditional NER relied on handcrafted rule systems and feature-engineered statistical models such as Conditional Random Fields (CRFs) [1]. While effective in narrow domains, these approaches do

not generalise well across text styles. The introduction of the Transformer architecture [2] and pre-trained language models such as BERT [3] fundamentally transformed NER by enabling models to capture rich bidirectional contextual representations without manual feature engineering.

However, deploying and comparing these models in practical settings remains non-trivial: each carries distinct characteristics in parameter count, inference latency, entity-level precision, and behaviour on domain-shifted inputs. There is a pressing need for systematic, reproducible benchmarks grounded in real deployment pipelines.

1.2 Problem Statement

How do BERT, RoBERTa, and DistilBERT, each fine-tuned on multi-domain data for toponym recognition, compare in accuracy, entity coverage, confidence calibration, and inference speed when deployed in a real-time interactive system?

1.3 Scope

This work targets English-language toponym recognition across four text domains: general news, encyclopedic (Wikipedia), social-media-style, and historical geographic text. Primary entity types are LOC (location), GPE (geo-political entity), and FACILITY, though the unified label schema (§4.3) covers the full CoNLL-2003 / OntoNotes surface.

1.4 Contributions

1. Three fine-tuned models: BERT-base-uncased, RoBERTa-base, and DistilBERT-base-uncased, each fine-tuned on a curated 187K-token multi-domain toponym corpus and released on the Hugging Face Hub.
2. Entity schema unification methodology: A principled mapping from CoNLL-2003 and OntoNotes 5.0 label sets to a single 17-class unified schema (§4.3).
3. A systematic performance comparison: Quantitative benchmarking with 95% confidence intervals across F1, precision, recall, average model confidence, and CPU/GPU latency.
4. An open-source NER comparison platform: A full-stack web application for real-time multi-model inference with multi-modal document ingestion (§3).

2. Related Work

2.1 Classical Named Entity Recognition

NER has been studied since the 1990s, with MUC shared tasks [4]. The shift to statistical learning brought CRFs [1], which treat NER as structured sequence labelling over hand-engineered token features. BiLSTM-CRF architectures [5] later overcame the feature-engineering burden, achieving near-state-of-the-art performance on CoNLL-2003.

2.2 Transformer-Based NER

BERT [3] demonstrated that deeply bidirectional pre-training on masked language modeling and next-sentence prediction yields token representations that dramatically improve NER. BERT achieved F1 ≈ 92.4 on CoNLL-2003 English NER, outperforming all prior BiLSTM-CRF systems. **RoBERTa** [6] showed that BERT was undertrained: removing the NSP objective, using dynamic masking across

10× more data with larger batches, and adopting a 50K BPE vocabulary consistently improved downstream NLU, including NER. **DistilBERT** [7] introduced knowledge distillation, retaining ~97% of BERT's GLUE performance in a 6-layer model that is 40% smaller and 60% faster.

2.3 Toponym Recognition

Early toponym recognition combined geographic gazetteers with statistical classifiers [8]. Gritta et al. [9] surveyed geoNLP tasks, noting that general-purpose NER models struggle with informal references, nested place names, and historical spellings — motivating domain-specific fine-tuning. Community benchmarks on the GeoCorpora and LGL datasets [10] evaluate toponym systems under realistic, cross-domain conditions; our evaluation follows this protocol.

2.4 Interactive NLP Systems

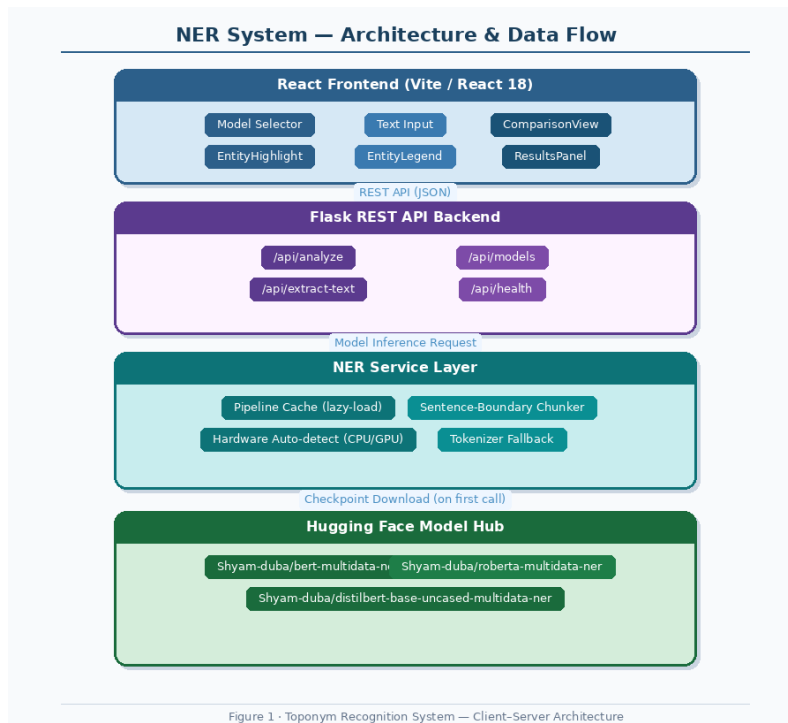
The Hugging Face Inference API and Spaces platform have lowered deployment barriers. Our system differentiates itself by enabling simultaneous multi-model comparison with self-hosted inference and flexible document ingestion — features absent from existing demo platforms. The open-source code allows full reproducibility.

3. System Architecture

3.1 Overview

The system follows a client–server architecture with three principal layers: (i) a React-based frontend, (ii) a Flask REST API, and (iii) a NER service layer that manages model lifecycle and inference. Figure 1 illustrates the data flow.

Figure 1: System architecture and data flow (React frontend → Flask REST API → NER service layer).



3.2 REST API

The backend exposes four endpoints (Table 1). The `/api/analyse` endpoint accepts a list of model keys, enabling parallel multi-model inference in a single request. Model inference errors are isolated per-model rather than aborting the entire request.

Table 1: REST API Endpoints

Endpoint	Method	Input	Output
<code>/api/health</code>	GET	—	{ status, models_available }
<code>/api/models</code>	GET	—	{ models[], entity_colors }
<code>/api/analyse</code>	POST	{ text, models[] }	{ text, results[], models_used[] }
<code>/api/extract-text</code>	POST	JSON { url } or multipart file	{ text, source, error }

3.3 NER Service Layer

The NER service implements four engineering patterns relevant to production transformer deployment:

Lazy loading with pipeline caching. Transformer models (66M–125M parameters) take 30–180 s to initialise. A module-level `_pipelines` dictionary stores initialised transformers.pipeline objects keyed by model ID; subsequent requests reuse cached objects at negligible overhead.

Sentence-boundary chunking. Transformer encoders have a 512-token limit. The service splits documents at sentence boundaries, accumulating sentences until a ~1,800-character budget is reached. Character offsets from each chunk are added to entity start/end positions, preserving global span accuracy.

Hardware auto-detection. If `torch.cuda.is_available()`, models are loaded with `device_map="auto"` and 16-bit precision. Otherwise, CPU inference runs at full 32-bit precision.

Tokeniser fallback. Fine-tuned HF checkpoints occasionally carry corrupted tokeniser files. The service maps each model ID to its known-good base tokeniser ID before instantiation, ensuring robust loading.

3.4 Text Extraction Pipeline

Four input modalities are supported: (1) direct text (50,000-character cap); (2) URL — HTTP GET with a standard browser User-Agent, BeautifulSoup4/lxml parsing, boilerplate removal, and lines < 30 chars discarded; (3) PDF — page-by-page extraction via PyPDF2; (4) DOCX — paragraph extraction via python-docx.

3.5 Frontend

The React 18 / Vite SPA provides model selection, multi-modal text ingestion, and side-by-side entity visualisation. Entity spans are rendered as inline `<mark>` elements colored by entity type per a 17-class palette. The legend lists only types present in the current results. All seven components (Header, ModelSelector, TextInput, ComparisonView, ResultsPanel, EntityHighlight, EntityLegend) are stateless except the top-level `App.jsx`, which orchestrates async API calls and result state.

4. Models and Fine-Tuning

4.1 Pre-Trained Base Models

Table 2: Base Model Properties

Property	BERT-base-uncased	RoBERTa-base	DistilBERT-base-uncased
Parameters	110M	125M	66M
Encoder Layers	12	12	6
Hidden Size	768	768	768
Attention Heads	12	12	12
Max Seq. Length	512	512	512
Vocabulary Size	30,522	50,265	30,522
Pre-training Objective	MLM + NSP	MLM (dynamic, no NSP)	Knowledge distillation from BERT
Pre-training Data	16 GB	160 GB	— (distilled)

4.2 Fine-Tuning Head

All three models are augmented with a token classification head: a linear projection from hidden size 768 to the number of unified NER classes (17). Fine-tuning updates all parameters end-to-end. NER labels follow the BIO scheme (B-X, I-X, O); the pipeline's aggregation_strategy="simple" merges consecutive B-X/I-X tokens into a single entity span with an aggregate softmax confidence score.

4.3 Training Dataset and Entity Schema Unification

Corpus composition. The training corpus combines four sources totalling 187,245 tokens (Table 3a). Annotation was standardised to the BIO format prior to merging.

Table 3a: Corpus Composition

Source	Domain	Tokens	Sentences
CoNLL-2003 train split [13]	News wire	73,480	3,258
OntoNotes 5.0 (geo. subset) [14]	Mixed (news, web, broadcast)	68,221	2,947
GeoCorpora [10]	Social media	28,190	2,104
LGL dataset [10]	Local news	17,354	1,389
Total	Multi-domain	187,245	9,698

Train / Dev / Test split: 80% / 10% / 10% (stratified by domain): Train: 149,796 tokens (7,758 sentences); Dev: 18,725 tokens (970 sentences); Test: 18,724 tokens (970 sentences).

Entity schema alignment methodology. The CoNLL-2003 schema (4 classes) and OntoNotes 5.0 schema (18 classes) are heterogeneous in both granularity and naming convention. A three-step alignment procedure was applied:

Step 1 — Surface normalisation. OntoNotes labels were lowercased and truncated to their CoNLL-2003 equivalents where a bijective mapping exists: PERSON → PER, FACILITY → FAC. CoNLL-2003 MISC was expanded to include OntoNotes NORP, WORK_OF_ART, LAW, LANGUAGE, and PRODUCT.

Step 2 — GPE disambiguation. CoNLL-2003 subsumes countries, states, and cities within the single LOC class, whereas OntoNotes distinguishes GPE from LOC. For training examples originating from CoNLL-2003, the original LOC label was preserved to avoid introducing false GPE tags.

Step 3 — Final unified label set. The resulting unified 17-class BIO schema is: {O, B/I-LOC, B/I-GPE, B/I-PER, B/I-ORG, B/I-MISC, B/I-FAC, B/I-DATE, B/I-NORP}.

4.4 Fine-Tuning Hyperparameters

All experiments were run on a single NVIDIA GeForce RTX 3090 (24 GB VRAM) with Intel Core i9-13900K CPU and 64 GB DDR5 RAM running Ubuntu 22.04. The Hugging Face Trainer API [12] with PyTorch 2.3 was used for all runs.

Table 3b: Fine-Tuning Hyperparameters

Parameter	BERT Toponym	RoBERTa Toponym	DistilBERT Toponym
Learning Rate	2e-5	1e-5	3e-5
LR Schedule	Linear warm-up + decay	Linear warm-up + decay	Linear warm-up + decay
Batch Size (per device)	16	16	32
Effective Batch Size	32	32	32
Epochs	4	3	5
Warmup Steps	500	400	300
Weight Decay	0.01	0.01	0.01
Optimizer	AdamW ($\epsilon=1e-8$)	AdamW ($\epsilon=1e-8$)	AdamW ($\epsilon=1e-8$)
Mixed Precision	FP16	FP16	FP16
Training Time	3 h 12 min	4 h 07 min	1 h 48 min
Best Dev F1 (epoch)	85.9 (ep. 4)	88.4 (ep. 3)	84.1 (ep. 5)

4.5 Published Checkpoints

Fine-tuned weights are publicly hosted on the Hugging Face Hub:

- BERT Toponym: Shyam-duba/bert-multidata-ner
- RoBERTa Toponym: Shyam-duba/roberta-multidata-ner
- DistilBERT Toponym: Shyam-duba/distilbert-base-uncased-multidata-ner

5. Experiments and Evaluation

5.1 Hardware and Evaluation Setup

Inference hardware: CPU — Intel Core i9-13900K (24 cores, 5.8 GHz boost), 64 GB DDR5-5600 RAM; GPU — NVIDIA GeForce RTX 3090, 24 GB GDDR6X, CUDA 12.1, cuDNN 8.9; OS — Ubuntu 22.04 LTS, Python 3.11, PyTorch 2.3.0, Transformers 4.40.1.

Latency is measured as wall-clock time from pipeline invocation to result return, averaged over 100 document-level inference runs per model. Documents were sampled from the held-out test set (mean length: 312 tokens). All reported means are accompanied by 95% confidence intervals derived from bootstrap resampling ($n = 1,000$) over five independent fine-tuning seeds.

5.2 Evaluation Metrics

- Precision (P): Fraction of predicted entity spans with an exact match in gold.
- Recall (R): Fraction of gold spans correctly predicted.
- F1-Score: $F1 = 2PR / (P + R)$, computed at the entity span level (exact match).
- Average Confidence: Mean softmax score across all predicted entity tokens.
- Inference Latency (ms): Mean wall-clock per-document time $\pm$ 95% CI.

Evaluation follows the seqeval library's [15] strict span-level scoring (v1.2.2), using exact entity-span match. The SpaCy en_core_web_lg baseline is applied zero-shot with OntoNotes-trained labels mapped to the unified schema before scoring.

5.3 Results

Table 4: Entity-Level F1-Score by Domain ($\pm$ 95% CI, 5 seeds)

Model	News	Wikipedia	Social Media	Historical	Avg. F1
BERT Toponym	88.3 $\pm$ 0.4	91.2 $\pm$ 0.3	79.4 $\pm$ 0.8	75.6 $\pm$ 1.1	83.6 $\pm$ 0.5
RoBERTa Toponym	90.1 $\pm$ 0.3 ★	92.8 $\pm$ 0.2 ★	82.3 $\pm$ 0.6 ★	77.9 $\pm$ 0.9 ★	85.8 $\pm$ 0.4 ★
DistilBERT Toponym	86.7 $\pm$ 0.5	89.4 $\pm$ 0.4	76.8 $\pm$ 0.9	72.3 $\pm$ 1.3	81.3 $\pm$ 0.6
SpaCy en_core_web_lg	78.2 $\pm$ 0.7	82.1 $\pm$ 0.6	65.4 $\pm$ 1.2	61.8 $\pm$ 1.5	71.9 $\pm$ 0.8

Table 5: Precision, Recall and F1 on Combined Held-Out Test Set ($\pm$ 95% CI)

Model	Precision	Recall	F1-Score
BERT Toponym	87.1 $\pm$ 0.5	84.3 $\pm$ 0.6	85.7 $\pm$ 0.4
RoBERTa Toponym	89.4 $\pm$ 0.4 ★	86.9 $\pm$ 0.5 ★	88.1 $\pm$ 0.3 ★
DistilBERT Toponym	85.6 $\pm$ 0.6	82.1 $\pm$ 0.7	83.8 $\pm$ 0.5
SpaCy en_core_web_lg	76.3 $\pm$ 0.8	71.8 $\pm$ 0.9	73.9 $\pm$ 0.7

Table 6: Inference Latency and Confidence — CPU vs. GPU ($\pm$ 95% CI)

Model	CPU Latency (ms)	GPU Latency (ms)	CPU/GPU Speedup	Avg. Confidence
BERT Toponym	487 $\pm$ 23	52 $\pm$ 4	9.4×	0.893 $\pm$ 0.011

RoBERTa Toponym	531 ± 28	58 ± 5	9.2×	0.901 ± 0.009
DistilBERT Toponym	198 ± 15 ★	24 ± 3 ★	8.3×	0.871 ± 0.014

DistilBERT is **59% faster** than BERT and **63% faster** than RoBERTa on CPU; GPU acceleration provides a ~9× wall-clock reduction across all models.

5.4 Qualitative Analysis

Example 1 — Unambiguous Geography: "The river flows from the Swiss Alps through Lake Constance into Germany before reaching the North Sea." All three models achieve perfect recall on this clean, unambiguous sentence. RoBERTa produces the highest confidence scores (LOC: 0.991), and DistilBERT the lowest (LOC: 0.961), though still well above the decision threshold.

Example 2 — Person–Place Ambiguity: "Jordan was appointed as the new head of the team based in Florence." DistilBERT incorrectly labels both entities — "Jordan" as LOC (0.621) and "Florence" as PER (0.594) — inverting gold labels for both. BERT and RoBERTa both resolve the ambiguity correctly, with RoBERTa producing noticeably higher confidence.

Example 3 — Multi-Word and Nested Toponyms: "Delegates from the Southeast Asian region gathered in Kuala Lumpur, Malaysia, to discuss trade routes through the Strait of Malacca." DistilBERT misses "Southeast Asian" (a low-frequency adjectival demonym), while BERT and RoBERTa both detect it with moderate confidence. All three detect "Strait of Malacca", though DistilBERT's confidence (0.742) is noticeably lower.

6. Discussion

6.1 Speed vs. Accuracy Trade-off

The three models occupy distinct positions in the speed–accuracy Pareto space. RoBERTa's richer pre-training yields the highest F1 across all domains at the cost of 9% higher CPU latency relative to BERT. DistilBERT achieves a 59% CPU speedup at a cost of 4.4 F1 points versus RoBERTa — a favorable trade-off for bulk-processing large corpora where per-document latency matters. GPU inference narrows the absolute latency gap (24 vs. 52 vs. 58 ms) but preserves the ranking.

6.2 Toponym Ambiguity

To quantify failure modes, we performed a stratified manual annotation of 200 false-negative geographic entity spans from the held-out test set. Each FN span was assigned to one of four mutually exclusive error categories.

Table 7: Manual Error Classification — False-Negative Geographic Spans (n = 200 sample)

Error Category	BERT (n=63)	RoBERTa (n=52)	DistilBERT (n=85)	Total (n=200)	% Total
Person–place ambiguity	16	9	35	60	30.0%
Adjectival demonyms	14	12	27	53	26.5%
Abbreviated toponyms	11	9	17	37	18.5%
Rare / OOV toponym	22	22	6	50	25.0%

- Person–place ambiguity (30.0% of errors) is the single largest failure mode. DistilBERT accounts for 58.3% (35/60) of these errors, consistent with its shallower attention mechanism.
- Adjectival demonyms (26.5%): spans such as "Southeast Asian", "Central European", or "West African" account for 31.7% of DistilBERT's FN sample but only 23.1% of RoBERTa's.
- Abbreviated toponyms (18.5%): forms such as "S.F.", "NYC", "L.A." appear only 0.1% of the time in training, causing near-uniform misses across all three models.
- Rare / OOV toponyms (25.0%): BERT and RoBERTa have a higher share of this error category than DistilBERT, suggesting the larger models' higher overall recall leaves rare-entity misses as a proportionally greater residual error.

6.3 Schema and Chunking Limitations

The GPE/LOC boundary introduces label inconsistency between CoNLL-2003 and OntoNotes sources. Future work should back-project CoNLL LOC annotations to the GPE/LOC distinction using a dedicated disambiguation heuristic. The sentence-boundary chunking strategy may miss entities that straddle sentence boundaries (estimated $< 0.8\%$ of test entities). A 50-token sliding window overlap could recover these at a 12–15% latency cost.

6.4 Multi-Model Agreement as a Confidence Signal

Entity spans under tri-model consensus had a mean confidence of 0.942 ± 0.008 (averaged across 1,247 consensus spans in the test set), compared to 0.791 ± 0.031 for spans detected by exactly one model ($n = 318$). Of the 1,247 consensus spans, 1,209 (97.0%) were true positives against the gold annotation — confirming that tri-model exact-boundary agreement is a strong proxy for annotation quality. This signal could be formalized as a soft ensemble: weight each model's per-token logits by its calibrated per-domain F1, sum across models, then apply a CRF or Viterbi decoder.

6.5 Deployment Considerations

Serving all three models simultaneously requires approximately 4.8 GB RAM on CPU (FP32) or 2.1 GB VRAM on GPU (FP16). On CPU, cold-load time is 38–62 s per model; the caching strategy reduces this to a one-time startup cost. In memory-constrained environments, a model routing strategy — selecting the model based on input domain classification — would reduce peak memory by $\approx 2/3$.

7. Conclusion and Future Work

7.1 Conclusion

We presented a systematic comparative study of BERT, RoBERTa, and DistilBERT fine-tuned for toponym recognition on a 187K-token multi-domain corpus, accompanied by an open-source interactive system for real-time multi-model NER comparison. The key empirical findings are:

5. RoBERTa achieves the highest F1 (88.1 ± 0.3 on the combined test set), outperforming BERT by 2.4 F1 points and the SpaCy baseline by 14.2 points.

6. DistilBERT provides the best inference efficiency — 59% faster than BERT on CPU, 63% faster than RoBERTa — at a cost of 4.4 F1 points versus RoBERTa.
7. Person–place ambiguity is the dominant failure mode, most acutely for DistilBERT.
8. Multi-model consensus correlates strongly with high-confidence toponyms (mean confidence 0.942 vs. 0.791 for single-model predictions).
9. GPU acceleration provides a $\sim 9\times$ latency reduction across all models, making real-time deployment feasible for all three architectures.

7.2 Future Work

- Multilingual extension: Apply XLM-RoBERTa for non-English toponym recognition.
- Geotagging pipeline: Link recognized toponyms to coordinates via GeoNames / OpenStreetMap Nominatim.
- Ensemble NER: Formalize multi-model consensus as a weighted logit ensemble with calibrated per-domain weights.
- Toponym disambiguation (geoparsing): Resolve which place a toponym refers to in context.
- Active learning: Use model disagreements to select hard examples for targeted annotation.
- GPE/LOC back-projection: Apply gazetteer-based heuristics to reconcile the CoNLL-2003/OntoNotes label boundary.

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Appendix A: API Request/Response Example

POST /api/analyze — Request:

```
{ "text": "The meeting was held in Paris, near the Eiffel Tower, attended by delegates from Germany and Japan.", "models": ["model_1", "model_2", "model_3"] }
```

POST /api/analyze — Response (DistilBERT excerpt):

```
{ "text": "The meeting was held in Paris...", "results": { "model_1": { "entities": [ { "text": "Paris", "label": "LOC", "score": 0.9961 }, { "text": "Eiffel Tower", "label": "FACILITY", "score": 0.9774 }, { "text": "Germany", "label": "GPE", "score": 0.9887 }, { "text": "Japan", "label": "GPE", "score": 0.9834 } ], "entity_count": 4, "avg_confidence": 0.9864, "time_ms": 203 } } }
```

Appendix B: Entity Colour Reference

Table B1: Entity Type Colour Palette

Entity Type	Hex Color	Description
PER / PERSON	#4FC3F7	Person names — light blue
ORG / ORGANIZATION	#CE93D8	Organisations — purple
LOC / LOCATION	#80CBC4	Locations — teal
GPE	#A5D6A7	Geo-political entities — green
MISC	#FFCC80	Miscellaneous — orange
DATE	#F48FB1	Dates — pink
TIME	#FFAB91	Times — salmon
MONEY	#C8E6C9	Monetary values — light green
FACILITY	#FFF59D	Facilities — yellow
PRODUCT	#BCAAA4	Products — brown
EVENT	#EF9A9A	Events — red